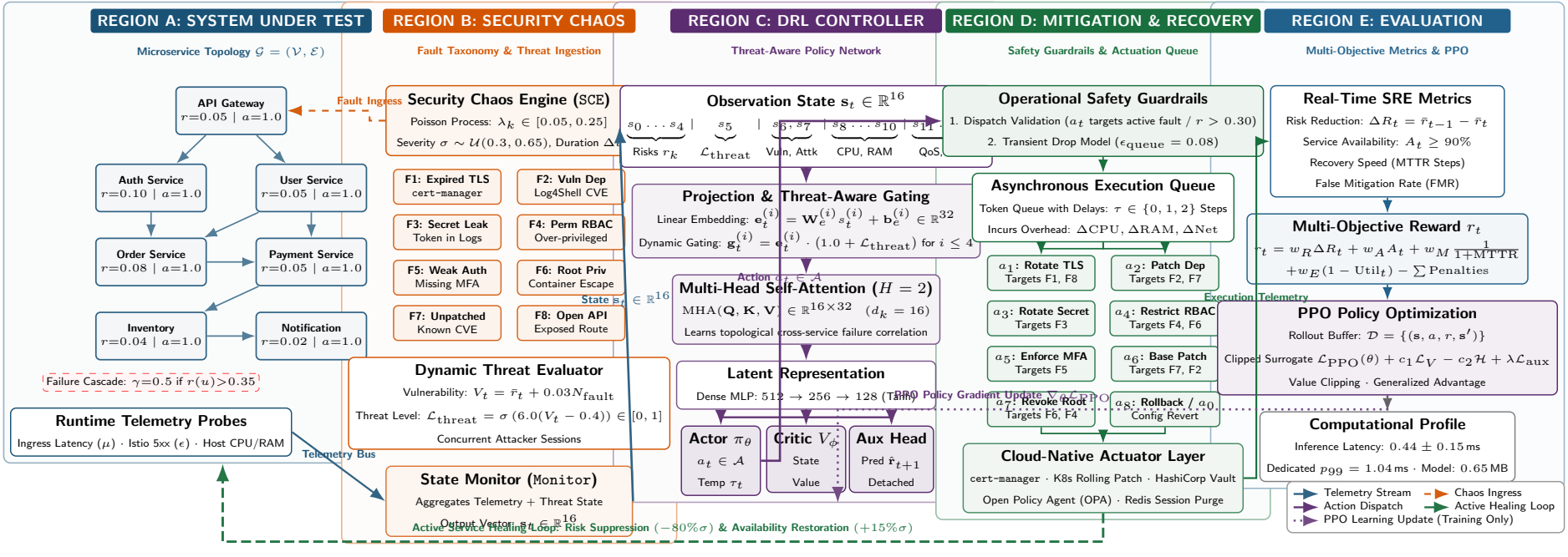

REGION A: SYSTEM UNDER TEST
Microservice Topology $\mathcal{G} = (\mathcal{V}, \mathcal{E})$
API Gateway
r=0.05 | a=1.0
Auth Service
r=0.10 | a=1.0
User Service
r=0.05 | a=1.0
Order Service
r=0.08 | a=1.0
Payment Service
r=0.05 | a=1.0
Inventory
r=0.04 | a=1.0
Notification
r=0.02 | a=1.0
Failure Cascade: γ=0.5 if $r(u)$>0.35
Runtime Telemetry Probes
Ingress Latency (μ) · Istio 5xx (ϵ) · Host CPU/RAM
Telemetry Bus
REGION B: SECURITY CHAOS
Fault Taxonomy & Threat Ingestion
Fault Ingress
Security Chaos Engine (SCE)
Poisson Process: $\lambda_k \in [0.05, 0.25]$
Severity $\sigma \sim \mathcal{U}(0.3, 0.65)$, Duration Δ
F1: Expired TLS
cert-manager
F2: Vuln Dep
Log4Shell CVE
F3: Secret Leak
Token in Logs
F4: Perm RBAC
Over-privileged
F5: Weak Auth
Missing MFA
F6: Root Priv
Container Escape
F7: Unpatched
Known CVE
F8: Open API
Exposed Route
Dynamic Threat Evaluator
Vulnerability: $V_t = \bar{r}_t + 0.03 N_{\text{fault}}$
Threat Level: $\mathcal{L}_{\text{threat}} = \sigma\,(6.0(V_t - 0.4)) \in [0, 1]$
Concurrent Attacker Sessions
State Monitor (Monitor)
Aggregates Telemetry + Threat State
Output Vector: $\mathbf{s}_n \in \mathbb{R}^{16}$
REGION C: DRL CONTROLLER
Threat-Aware Policy Network
Observation State $\mathbf{s}_t \in \mathbb{R}^{16}$
$s_0 \dots s_4$ | s_5 | s_6, s_7 | $s_8 \dots s_{10}$ | s_{11} ...
Risks r_k
$\mathcal{L}_{\text{threat}}$
Vuln, Attk
CPU, RAM
QoS,
Projection & Threat-Aware Gating
Linear Embedding: $\mathbf{e}_t^{(i)} = \mathbf{W}_e^{(i)} s_t^{(i)} + \mathbf{b}_e^{(i)} \in \mathbb{R}^{32}$
Dynamic Gating: $\mathbf{g}_t^{(i)} = \mathbf{e}_t^{(i)} \cdot (1.0 + \mathcal{L}_{\text{threat}})$ for $i \leq 4$
Action $a_t \in \mathcal{A}$
State $\mathbf{s}_t \in \mathbb{R}^{16}$
Multi-Head Self-Attention ($H = 2$)
$\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \in \mathbb{R}^{16 \times 32}$ $(d_k = 16)$
Learns topological cross-service failure correlation
Latent Representation
Dense MLP: 512 → 256 → 128 (Tanh)
Actor π_θ
$a_t \in \mathcal{A}$
Temp τ_t
Critic V_ϕ
State
Value
Aux Head
Pred $\hat{\mathbf{r}}_{t+1}$
Detached
REGION D: MITIGATION & RECOVERY
Safety Guardrails & Actuation Queue
Operational Safety Guardrails
1. Dispatch Validation (a_t targets active fault / $r > 0.30$)
2. Transient Drop Model ($\epsilon_{\text{queue}} = 0.08$)
Asynchronous Execution Queue
Token Queue with Delays: $\tau \in \{0, 1, 2\}$ Steps
Incurs Overhead: ΔCPU, ΔRAM, ΔNet
a_1: Rotate TLS
Targets F1, F8
a_2: Patch Dep
Targets F2, F7
a_3: Rotate Secret
Targets F3
a_4: Restrict RBAC
Targets F4, F6
a_5: Enforce MFA
Targets F5
a_6: Base Patch
Targets F7, F2
a_7: Revoke Root
Targets F6, F4
a_8: Rollback / a_0
Config Revert
Cloud-Native Actuator Layer
cert-manager · K8s Rolling Patch · HashiCorp Vault
Open Policy Agent (OPA) · Redis Session Purge
REGION E: EVALUATION
Multi-Objective Metrics & PPO
Real-Time SRE Metrics
Risk Reduction: $\Delta R_t = \bar{r}_{t-1} - \bar{r}_t$
Service Availability: $A_t \geq 90\%$
Recovery Speed (MTTR Steps)
False Mitigation Rate (FMR)
Multi-Objective Reward r_t
$r_t = w_R \Delta R_t + w_A A_t + w_M \frac{1}{1+\text{MTTR}}$
$+ w_E(1 - \text{Util}_t) - \sum \text{Penalties}$
Execution Telemetry
PPO Policy Optimization
Rollout Buffer: $\mathcal{D} = \{(\mathbf{s}, a, r, \mathbf{s}')\}$
Clipped Surrogate $\mathcal{L}_{\text{PPO}}(\theta) + c_1 \mathcal{L}_V - c_2 \mathcal{H} + \lambda \mathcal{L}_{\text{aux}}$
Value Clipping · Generalized Advantage
PPO Policy Gradient Update $\nabla_\theta \mathcal{L}_{\text{PPO}}$
Computational Profile
Inference Latency: 0.44 ± 0.15 ms
Dedicated $p_{99} = 1.04$ ms · Model: 0.65 MB
Active Service Healing Loop: Risk Suppression ($-80\%\sigma$) & Availability Restoration ($+15\%\sigma$)
Telemetry Stream
Action Dispatch
PPO Learning Update (Training Only)
Chaos Ingress
Active Healing Loop